Proof. First, since X is reflexive and $\Psi(A)$ is power-bounded on X, the Uniform Mean Ergodic Theorem (see, e.g., [14] or [6]) implies that the Cesàro averages $N^{-1}\sum_{n=0}^{N-1}(\Psi(A))^n$ converge in the strong operator topology to a projection P (the *mean ergodic projection*) satisfying $\operatorname{Ran} P = \ker(I-\Psi(A))$ and $P\,\Psi(A) = \Psi(A)\,P = P$ [16, 17]. This yields part (a). Note that $\Psi(A)$ commutes with A by construction, and since P is a limit of polynomials in $\Psi(A)$ (which all commute with A), we deduce $PA = AP$.

For part (b), assume $\sigma(\Psi(A)) \cap \mathbb{T} \subset \{1\}$ and that 1 is an isolated spectral value of $\Psi(A)$ which is a pole of the resolvent (this is the case, for example, if 1 is an eigenvalue of $\Psi(A)$ of finite algebraic multiplicity and there are no other spectrum points on the unit circle) [18, 19]. Then the Katznelson–Tzafriri theorem and the Arendt–Batty–Lyubich–Vũ theorem on stability of powers (see [7, 8], or e.g. [20]) imply that $(\Psi(A))^n \to P$ strongly as $n \to \infty$ [20, 19]. (In fact, these theorems assert that $\Psi(A)^n x \to Px$ for all $x \in X$ whenever $\sigma(\Psi(A)) \cap \mathbb{T} \subseteq \{1\}$ and either 1 is a pole of the resolvent or at least a simple pole with no other spectrum on the unit circle.) This yields strong convergence $A^{(n)} = (\Psi(A))^n \to P$. The proof of (b) is complete. □

Remark 3.2 (Range identification and joint fixed-points). Let us consider the range of the projection P obtained above. In general, from part (a) we only know $\operatorname{Ran} P = \ker(I-\Psi(A)) = \{x : \Psi(A)x = x\}$. However, under an additional *separation hypothesis*, one can show that this space of fixed points coincides with the intersection of the fixed-point spaces of each layer $\varphi_k(A)$. Recall $\Phi_k(A) := \varphi_k(A)$, so $\ker(I-\Phi_k(A)) = \{x : \varphi_k(A)x = x\}$ is the set of vectors invariant under the k-th transform. Clearly, if x is fixed by each $\Phi_k(A)$, then it is fixed by $\Psi(A) = \Phi_K(A)\cdots\Phi_1(A)$. The converse containment need not hold in general; it holds, for example, if the condition

$$\forall \lambda \in \sigma(A),\ \Psi(\lambda) = \lambda \implies \varphi_k(\lambda) = \lambda \text{ for each } k, \tag{1}$$

is satisfied. In words, whenever the composite function Ψ has λ as a fixed point, each individual φ_k must also fix λ. In that case, the joint zero set of the functions $(\varphi_k(z) - z)$ coincides with the zero set of $(\Psi(z) - z)$ in the spectral domain. This implies (via a functional calculus argument using a factorization of $\Psi - I$) that the fixed subspace $\ker(I-\Psi(A))$ coincides with $\bigcap_{k=1}^{K}\ker(I-\Phi_k(A))$. We formalize this as Lemma 3.3 below, which is essentially Lemma 1.2 of the original manuscript [15, 16].

Lemma 3.3 (Separation at the fixed value)**.** *With A, φ_k, Ψ as in Theorem C, assume the implication (1) holds for all λ in the spectral domain of A. Then* $\ker(I-\Psi(A)) = \bigcap_{k=1}^{K}\ker(I-\varphi_k(A))$. *Equivalently,* $\operatorname{Ran} P = \{x : \Phi_k(A)x = x,\ \forall\, k = 1,\dots,K\}$.

Proof. We only need to show the inclusion $\ker(I-\Psi(A)) \subseteq \bigcap_k \ker(I-\varphi_k(A))$, since the opposite inclusion is immediate. Let $x \in X$ be such that $\Psi(A)x = x$. Consider the bounded holomorphic functions $\Psi - I$ and $\varphi_k - I$ on the spectral domain (disk or sector). By the spectral mapping principle (ensured by the functional calculus), the assumption (1) implies that the common zero-set of the functions $\{\varphi_k - I : k = 1,\dots,K\}$ coincides with the zero-set of $\Psi - I$. Since $\Psi - I = (\varphi_K - I)h$ for some holomorphic function h (one can factor out one of the factors $(\varphi_k - I)$, though not necessarily all of them, but this one-factor factorization is enough), the above means that $(\Psi - I)$ vanishes exactly on the same set as $(\varphi_K - I)(\varphi_{K-1} - I)\cdots(\varphi_1 - I)$. Therefore, in the (commuting) functional calculus of A, we have $\ker(\Psi(A) - I) = \ker((\varphi_K(A) - I)\cdots(\varphi_1(A) - I))$. However, since all $\varphi_k(A)$ commute (being functional images of a common operator A), the idempotent factors as the product of the individual Riesz projections $Q_k := E_A(\{\lambda : \varphi_k(\lambda) = \lambda\})$ (here we use that 1 is an isolated spectral point of each $\varphi_k(A)$ by assumption $\varphi_k(1) = 1$ and $|\varphi_k(\lambda)| < 1$ for other $\lambda \in \sigma(A)$). Namely, $\ker(\Psi(A) - I) = \ker(Q) = \ker(Q_1\cdots Q_K) = \ker(Q_1) \cap \cdots \cap \ker(Q_K) = \bigcap_k \ker(\varphi_k(A) - I)$; the inclusion $\supseteq$ is trivial, and $\subseteq$ holds because if $(\Psi(A) - I)x = 0$ and (for contradiction) $(\varphi_j(A) - I)x \neq 0$

for some j, then $(\Psi(A)-I)$ cannot vanish on the spectral support of x, since on that support $\varphi_j - I$ stays away from zero whereas $\Psi - I$ is a product of the $\varphi_k - I$). Thus $x \in \ker(\varphi_k(A) - I)$ for each k, i.e. x is fixed by each $\varphi_k(A)$. This establishes the desired inclusion. □

In the setting of Theorem C and Lemma 3.3, one expects condition (1) to hold under a mild version of the peripheral fixed-point property for each φ_k. Indeed, suppose λ is a spectral value on the boundary (say $|\lambda| = 1$ in the discrete case or $|\arg \lambda| = \Theta$ in the sectorial case). If $\Psi(\lambda) = \lambda$ but $\varphi_j(\lambda) \neq \lambda$ for some j, then that φ_j must have strictly decreased the modulus or moved λ inside, and the only way the composite Ψ returns to λ is through some compensating action of later transforms. This is intuitively impossible if each φ_k does not create new unimodular fixed points apart from those it inherited from λ. For example, if each φ_k is such that whenever $|\lambda| = 1$ and $|\varphi_k(\lambda)| = 1$ one must have $\varphi_k(\lambda) = \lambda$ (the same peripheral fixed-point property as in the normal case), then it follows that if $\Psi(\lambda) = \lambda$ (with $|\lambda| = 1$) one of the φ_k must have made $|\varphi_k(\lambda)| < 1$, which means all subsequent $\varphi_{k+1}, \ldots, \varphi_K$ output strictly inside the boundary, so $\Psi(\lambda)$ ends up with $|\Psi(\lambda)| < 1$, a contradiction. Thus $\varphi_k(\lambda)$ must equal λ for each k. Hence (1) holds for λ on the spectral boundary. For interior spectral values λ (those with $|\lambda| < 1$ or strictly inside sector), condition (1) may not hold automatically. However, if one assumes (as we did in the normal case analysis) that the only fixed point of each φ_k in the disk/sector is perhaps 0, and that any composite Ψ is not allowed to have some other interior fixed point appear, then effectively the only fixed points of Ψ are either in the set $\{\lambda : |\lambda| = 1, \varphi_k(\lambda) = \lambda \forall k\}$ or an interior Denjoy–Wolff point common to all. In the latter case, $\Psi(\lambda) = \lambda$ implies each $\varphi_k(\lambda) = \lambda$ because if a composition has an interior fixed, each map with that property usually shares it (this is more subtle but in many analytic families it holds, e.g. the first map having a different interior fixed would produce a second interior fixed for composite, improbable unless trivial). We will not pursue these technicalities further. In practice, one often proves that the only possible limit of the iterative process for any given initial λ is either 1 or 0, exactly as we did in Lemma 2.6 in the normal case. A similar approach can be applied in the Banach case using the Denjoy–Wolff theorem for holomorphic maps on the disk/sector. Under those conclusions, clearly P is the projection onto the peripheral spectral subspace of A. For brevity, we will not repeat the analogous argument here, as it closely parallels Lemma 2.6.

4 Examples and counterexamples

In this section we illustrate our results with several examples and point out the necessity of certain hypotheses via counterexamples.

Examples for normal operators (Hilbert space)

Throughout these examples, we take $A = M_a$ to be a normal operator given by multiplication by a measurable function $a(\xi)$ on some L^2-space. We assume $\sigma(A) \subseteq \overline{\mathbb{D}} = \{z : |z| \leq 1\}$. By applying Theorem 2.1 and Lemma .6 (dominated convergence) [4, 5], we only need to verify the pointwise convergence conditions on the spectral values; the operator convergence then follows.

Example 4.1 (Two-layer cycle with explicit closed form)**.** Let $p_1(z) = \frac{1}{2}(1+z^2)$ and $p_2(z) = \frac{z-1/2}{1-\frac{1}{2}z}$. These are two Schur functions (bounded by 1 on $\overline{\mathbb{D}}$) with $p_1(1) = 1$ and $p_2(1) = 1$. Consider the two-layer iteration with $f := p_2 \circ p_1$. Then one computes explicitly $f(z) = \frac{z^2}{2-z^2}$, so in fact $f(z) = \frac{1}{2}z^2(1-\frac{1}{2}z^2)^{-1}$. A calculation shows that $f^m(z) = \frac{(2/3)^m z^{2^m}}{1-(1-(2/3)^m)z^{2^m}}$ for $m \geq 1$. In the limit as $m \to \infty$, this rational expression converges to $\chi(z)$ given by $\chi(z) = 1_{\{1\}}(z)$. Indeed, if $z = 1$ then

$f^m(1) = 1$ for all m. If $z \neq 1$, then as $m \to \infty$, $(2/3)^m \to 0$ so $f^m(z) \to 0$. Thus $f^m(z) \to \chi(z)$ for all $z \in \overline{\mathbb{D}}$. Moreover, $\sup_m |f^m(z)| \leq 1$ for all $z \in \overline{\mathbb{D}}$. Therefore by Theorem 2.1 (or directly by the dominated convergence Lemma .6 [6, 7]), we conclude that for any normal operator A with $\sigma(A) \subseteq \overline{\mathbb{D}}$, $A^{(m)} \to E_A(\{1\})$. In particular, $A^{(m)} = f^m(A)$ converges strongly to the spectral projection of A onto $\{1\}$ (the part of the spectrum fixed by both p_1 and p_2). Note that in this example the stabilization only occurs in the limit stage $m \to \infty$, and no finite iterate $A^{(m)}$ is exactly a projection (though convergence takes place well before $m = \infty$ in the strong sense). Thus the transfinite stage $\alpha_* = \omega$ is attained; this example realizes the "extremal countable stage" $\alpha_* = \omega$ mentioned in the introduction.

Example 4.2 (Two-layer parametric cycle)**.** Fix a parameter $t \in (0, 1)$. Consider the transforms $p_1^{(t)}(z) = t + (1 - t)z$ and $p_2^{(t)}(z) = b_t(z) := \frac{z-t}{1-tz}$. Here $p_1^{(t)}$ is the affine map that sends 1 to 1 and 0 to t. The function $b_t(z)$ is a Blaschke factor (a disk automorphism) satisfying $b_t(t) = 0$ and $b_t(1) = 1$. Define $f_t := p_2^{(t)} \circ p_1^{(t)}$. Then $f_t(z) = \frac{z}{1+t-tz}$. Using a conjugation by the Möbius transform $T(z) = \frac{z}{1-z}$ (which maps $\mathbb{D}$ to the right half-plane and carries f_t to a dilation) [8, 9], one finds $T \circ f_t \circ T^{-1}(w) = \lambda w$ with $\lambda = \frac{1}{1+t}$. As $m \to \infty$, one has $\lambda^m \to 0$, so $f_t^m(z) = T^{-1}(\lambda^m T(z))$. For $z = 1$, note that $f_t^m(1) = 1$ for all m. If $z \neq 1$, then $f_t^m(z) = \frac{\lambda^m z}{1-(1-\lambda^m)z} \to 0$ as $m \to \infty$. The convergence is dominated by 1 as before, so for any normal A with $\sigma(A) \subseteq \overline{\mathbb{D}}$, $f_t^m(A) \to E_A(\{1\})$. This example shows that even a continuum of different two-layer transforms (one for each t) all lead to the same limiting projection onto the 1-eigenspace of A. Here $\Sigma_{\text{fix}} = \{1\}$ in all cases, but the convergence speeds vary with t. For instance, if t is close to 1, the iterate $p_1^{(t)}$ hardly moves points and the convergence is slow (requiring countably many steps in the transfinite sense); if t is near 0, convergence is more rapid.

Example 4.3 (Rational Schur family via an iterative Schur algorithm)**.** The previous example can be generalized by taking more complicated rational inner functions in the second layer. Specifically, one can construct a family of two-layer cycles $p_1^{(t)}(z) = t + (1 - t)z$ as before, and $p_{2,N}^{(t)}(z) = b_t(z)\Phi_N(b_t(z))$, where (Φ_N) is any sequence in the Schur class (bounded holomorphic functions on $\mathbb{D}$ with $\sup |\Phi_N| \leq 1$) that tends to 1 as $N \to \infty$ (locally uniformly in $\mathbb{D}$) [10, 11]. For example, one can take $\Phi_N(z) = (b_a(z))^N$ for some $a \in (0, 1)$, which yields $\Phi_N(z) \to 1$ as $N \to \infty$ on compacta away from $z = 1$. By applying the Schur algorithm (Appendix B) to $R(z) = p_{2,N}^{(t)} \circ p_1^{(t)}(z)$, one finds $R(z) = f_t(z)\Phi_N(f_t(z))$. As $N \to \infty$, this $R(z)$ converges to the function $f_t(z)$, which is exactly the two-layer cycle from Example 4.2. Moreover, any finite power $R^m(z)$ is also easy to analyze: since $p_{2,N}^{(t)}(1) = 1$ and $p_1^{(t)}(1) = 1$, it follows that $R^m(1) = 1$ for all m. If $z \neq 1$, then for each fixed m, as $N \to \infty$, $R^m(z) \to f_t^m(z)$ locally uniformly on $\mathbb{D}$. Thus for any fixed m, $R^m(z) \to \chi_m(z)$ pointwise, where χ_m is just f_t^m on $\sigma(A)$. But from Example 4.2 we know $f_t^m(z) \to \chi(z)$ (0 or 1) as $m \to \infty$. By a diagonal argument, one can let $N \to \infty$ sufficiently fast as $m \to \infty$ to see that $\lim_{m\to\infty}(p_{2,N_m}^{(t)} \circ p_1^{(t)})^m(z) = \chi(z)$ for every z. In this way, one obtains a more general family of two-layer cycles (with a parameter N as well as t) that still converge to the projection onto $\{1\}$. In fact, by choosing an appropriate $\{\Phi_N\}$, one can realize any rational inner function in the second layer. This shows that for normal A with $\sigma(A) \subseteq \mathbb{T} \cup \mathbb{D}$ (some part on unit circle and some inside), if one chooses any rational inner function Ψ that has $\Psi(1) = 1$ but $\Psi(\lambda) \neq \lambda$ for some spectral λ on $\mathbb{T}$ not equal to 1, then the iteration may not converge (indeed, the example of a rotation $\Psi(z) = e^{i\theta}z$ in the next subsection demonstrates divergence). Thus the peripheral fixed-point property is an essential hypothesis in Theorem 2.1 for ensuring convergence to a projection in the normal case.